Homework #1

Problem 1

Add comments to the following code and describe in one sentence what it computes. Assume that a0 is used for the input and initially contains n, a positive integer. Assume that a0 is used for the output.

(Set on Less than)

Code for Problem 1:

```
begin:  addi t0, x0, 0        // t0 = sum = 0
        addi t1, x0, 1        // t1=1
loop:   slt t2, a0, t1        // t2=(a0<t1)?1:0
        bne t2, x0, finish    // if(t2!=0) go to finish
        add t0, t0, t1        // t0=t0+t1
        addi t1, t1, 2        // t1=t1+2
        j loop                // Jump to Loop
finish: add a0, t0, x0        // a0=t0
```

计算所有小于等于n的奇数之和

Problem 2

Write a loop that reverses order of the bits of an 8-bit number in s0 and stores the result in s1. RISC-V registers have 4 bytes, but in this problem, the higher 24 bits of s0 are always 0. For example, if the lower 8 bits of s0 are 00001101, the lower 8 bites s1 should be 10110000.

```
begin:  addi t0, x0, 8
        addi s1, x0, 0
loop:   beq  t0, x0, finish
        andi t1, s0, 1
        slli s1, s1, 1
        add  s1, s1, t1
        srli s0, s0, 1
        addi t0, t0, -1
        jal  x0, loop
finish:
```

```
for(t0=8; t0>0; t0--){
  t1=s0&1
  s1<<1, s1=s1+t1
  s0>>1};
```

Problem 3

Assume we have an array in memory that contains int* arr = {1,2,3,4,5,6,0}. Let the values of arr be a multiple of 4 and stored in register s0. What do the snippets of RISC-V code do? Assume that all the instructions are run one after the other in the same context.

a) lw t0, 8(s0)

b) slli t1, t0, 2
add t2, s0, t1
lw t3, 0(t2)
addi t3, t3, 1
sw t3, 0(t2)

c) lw t0, 16(s0)
xori t0, t0, 0xFFF
addi t0, t0, 1

word

片段

t0 = arr[2] = 3
t1 = 4*t0 = 12
t2 = &arr + t1
t3 = [t2] = arr[3] = 4
t3 = t3 + 1 = 5
arr[3] = t3 = 5
t0 = arr[4] = 5
对t0按位取反
t0 = t0 + 1 = -5

arr[t0] += 1

00....0101
^ 11....1111
11....1010 = -6

Problem 4

Write a function sumSquare in RISC-V that, when given an integer n, returns the summation below. If n is not positive, then the function returns 0.

$$n^2 + (n-1)^2 + (n-2)^2 + \cdots + 1^2$$

For this problem, you are given a RISC-V function called square that takes in an integer and returns its square. Implement sumSquare using square as a subroutine.

```
sumSquare: addi sp, sp, -12
           sw   ra, 8(sp)
           sw   s0, 4(sp)
           sw   s1, 0(sp)

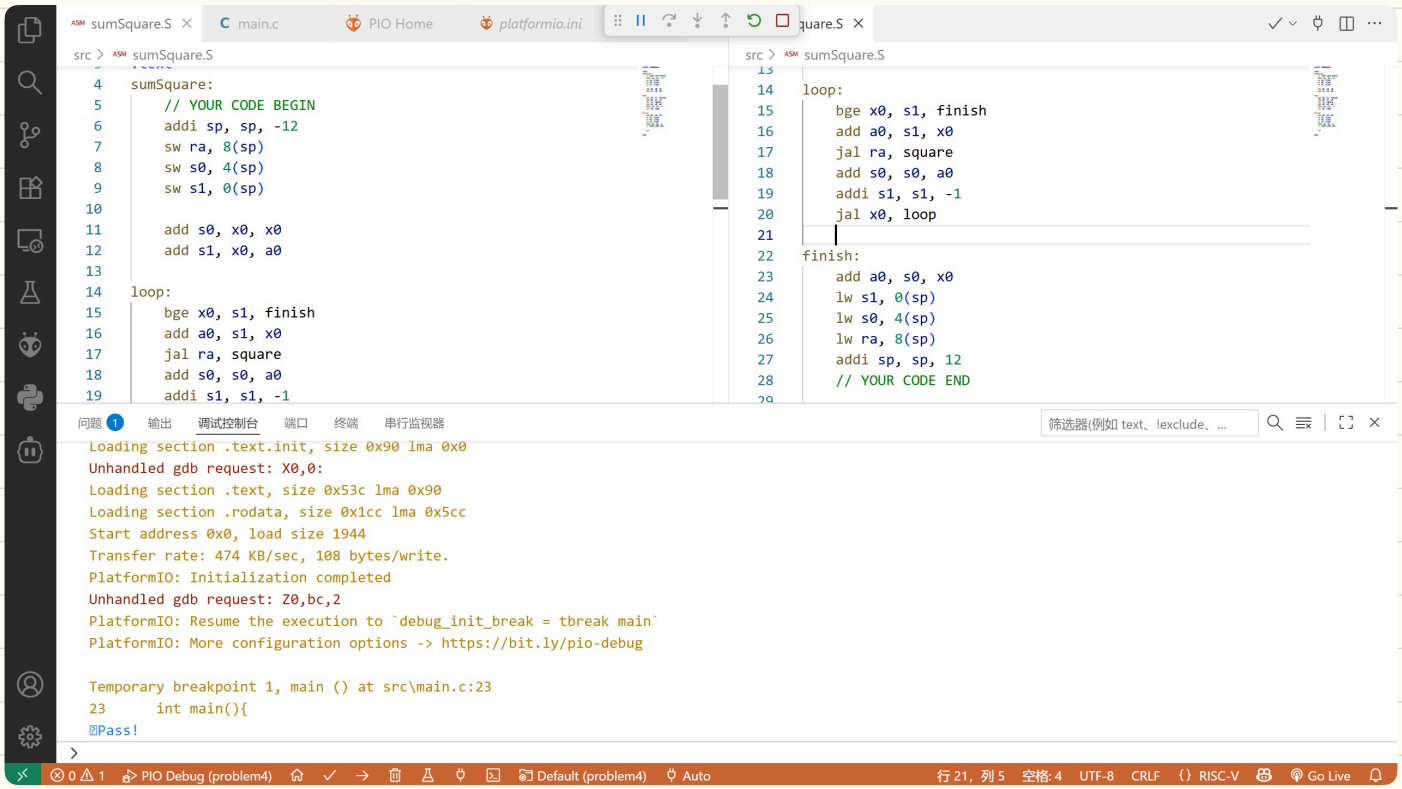
           add  s0, x0, x0
           add  s1, x0, a0

loop:      bge  x0, s1, finish
           add  a0, s1, x0
           jal  ra, square
           add  s0, s0, a0
           addi s1, s1, -1
           jal  x0, loop

finish:    add  a0, s0, x0
           lw   s1, 0(sp)
           lw   s0, 4(sp)
           lw   ra, 8(sp)
           addi sp, sp, 12
           jr   ra
```

```
for (s1=a0; s1>0; s1--)
{ s0+=square(s1) };
```

经调试，代码成功运行


sumSquare.S
main.c
PIO Home
platformio.ini
src > sumSquare.S
sumSquare:
// YOUR CODE BEGIN
addi sp, sp, -12
sw ra, 8(sp)
sw s0, 4(sp)
sw s1, 0(sp)
add s0, x0, x0
add s1, x0, a0
loop:
bge x0, s1, finish
add a0, s1, x0
jal ra, square
add s0, s0, a0
addi s1, s1, -1
jal x0, loop
finish:
add a0, s0, x0
lw s1, 0(sp)
lw s0, 4(sp)
lw ra, 8(sp)
addi sp, sp, 12
// YOUR CODE END
问题 1
输出
调试控制台
端口
终端
串行监视器
筛选器(例如 text、!exclude、...
Loading section .text.init, size 0x90 lma 0x0
Unhandled gdb request: X0,0:
Loading section .text, size 0x53c lma 0x90
Loading section .rodata, size 0x1cc lma 0x5cc
Start address 0x0, load size 1944
Transfer rate: 474 KB/sec, 108 bytes/write.
PlatformIO: Initialization completed
Unhandled gdb request: Z0,bc,2
PlatformIO: Resume the execution to `debug_init_break = tbreak main`
PlatformIO: More configuration options -> https://bit.ly/pio-debug
Temporary breakpoint 1, main () at src\main.c:23
23 int main(){
Pass!
PIO Debug (problem4)
Default (problem4)
Auto
行 21，列 5
空格: 4
UTF-8
CRLF
RISC-V
Go Live